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Inventor(s)

Chern; Jehn-Huar Howard et al.

CVD BORON UNIFORMITY OVERCOMING LOADING EFFECTS

Abstract

The loading effects of boron layers on silicon within a window may be reduced or eliminated by depositing an adhesion layer on a dielectric layer before depositing the boron layer. The adhesion layer may reduce or eliminate lateral diffusion of boron species into the window by being deposited on the adhesion layer. The approach using the adhesion layer may enable forming the boron layer at the nanometer scale within windows which are at the tens of millimeters scale and below. The boron layer and the silicon layer may form a detector which may be used in scanning electron microscopes and the like.

Inventors: Chern; Jehn-Huar Howard (Morgan Hill, CA), Trimpl; Marcel (San Jose, CA), Brown; David L. (Los Gatos, CA)

Applicant: KLA Corporation (Milpitas, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] The present application claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application Ser. No. 63/559,866, filed Feb. 29, 2024, titled “Improvement to CVD boron deposition uniformity due to loading effects”, which is incorporated herein by reference in the entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to solid-state semiconductor devices, and more particularly to electron and photon detectors for high-speed scanning in electron or optical microscopy applications.

BACKGROUND

[0003] Pure boron layers at nanometer-scale thicknesses may experience pattern dependency and loading effects when deposited on silicon within patterned windows. The deposition rate of pure boron using chemical vapor deposition (CVD) can vary over a wafer depending on the silicon opening size and oxide coverage ratio (OCR) of the mask design. The higher the oxide coverage ratio, the higher the deposition rate. The rate also increases as the sizes of the windows to the silicon decrease due to the local loading effect. The chemical vapor deposition process may experience a vertical diffusion from a boundary layer and a lateral diffusion along the oxide. Floating boron atoms laterally diffuse over the dielectric creating a higher concentration of boron atoms at the silicon-dielectric interface. Consequently, there is an increase of boron atoms sticking to the silicon surface in the proximity of the open windows. The loading effects may cause the thickness of the pure boron layers to vary over the width of the window and/or between multiple windows of different sizes. The variation in the thickness may render the pure boron layer ineffective for use in detectors as the width of the window decreases.

[0004] One method for reducing the localized loading effects is to decrease diborane partial pressure as well as carrier gas flow. The reduction of the pattern dependency with decreasing diborane partial pressure as well as carrier gas flow, have been applied to diodes with active areas on the order of hundreds of millimeters.

[0005] Some detector applications need high bandwidth detection, meaning the windows must be decreased. For high-speed diodes on the order of hundreds of MHz where drift times need to be optimized, small active areas are preferred, leading to a higher fraction of areas that are covered by dielectrics compared to the total area. Controlling the diborane partial pressure and the carrier gas flow may be inadequate to remedy the local loading effects for the small active areas. Therefore, it would be advantageous to provide a device, system, and method that cures the shortcomings described above.

SUMMARY

[0006] A detector is described in accordance with one or more embodiments of the present disclosure. In some embodiments, the detector may include: a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the

lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.

[0007] A method is described in accordance with one or more embodiments of the present disclosure. In some embodiments, the method may include: depositing an epitaxial layer on a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; depositing a dielectric layer on the epitaxial layer; depositing an adhesion layer on the dielectric layer; etching a window through the adhesion layer and the dielectric layer to the epitaxial layer; and depositing boron layers, the boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.

[0008] A scanning electron microscope is described in accordance with one or more embodiments of the present disclosure. In some embodiments, the scanning electron microscope may include: a detector including: a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.

[0009] A detector array is described, in accordance with one or more embodiments of the present disclosure. In some embodiments, the detector array may include an array of detectors, the array of detectors may include a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the silicon layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.

[0010] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not necessarily restrictive of the present disclosure. The accompanying drawings, which are incorporated in and constitute a part of the

specification, illustrate subject matter of the disclosure. Together, the description and drawings serve to explain the principles of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The numerous advantages of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures in which:

[0012] FIG. 1A depicts a cross-section view of a detector, in accordance with one or more embodiments of the present disclosure.

[0013] FIG. 1B depicts the cross-section view of the detector with a capping layer, in accordance with one or more embodiments of the present disclosure.

[0014] FIG. 2 depicts a flow diagram of a method of manufacturing the detector, in accordance with one or more embodiments of the present disclosure.

[0015] FIG. 3A depicts depositing a dielectric layer on an epitaxial layer, in accordance with one or more embodiments of the present disclosure.

[0016] FIG. 3B depicts depositing an adhesion layer on the dielectric layer, in accordance with one or more embodiments of the present disclosure.

[0017] FIG. 3C depicts etching a window through the adhesion layer and the dielectric layer to the epitaxial layer, in accordance with one or more embodiments of the present disclosure.

[0018] FIG. 3D depicts depositing boron layers on the epitaxial layer within the window and on the adhesion layer, in accordance with one or more embodiments of the present disclosure.

[0019] FIG. 3E depicts depositing a capping layer, in accordance with one or more embodiments of the present disclosure.

[0020] FIG. 3F depicts removing the upper boron layer and the adhesion layer, in accordance with one or more embodiments of the present disclosure.

[0021] FIG. 3G depicts the detector after a cathode and anode are formed by further processing, in accordance with one or more embodiments of the present disclosure.

[0022] FIG. 4A depicts a top view of the detector, in accordance with one or more embodiments of the present disclosure.

[0023] FIG. 4B depicts a cross-section view of the detector, in accordance with one or more embodiments of the present disclosure.

[0024] FIG. 4C depicts a top view of the detector configured with multiple channels, in accordance with one or more embodiments of the present disclosure.

[0025] FIG. 5 depicts a simplified block diagram of a scanning electron microscope, in accordance with one or more embodiments of the present disclosure.

[0026] FIG. 6 depicts a top view of a detector array, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] The present disclosure has been particularly shown and described with respect to certain embodiments and specific features thereof. The embodiments set forth herein are taken to be illustrative rather than limiting. It should be readily apparent to those of ordinary skill in the art that various changes and modifications in form and detail may be made without departing from the spirit and scope of the disclosure. Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings.

[0028] Embodiments of the present disclosure are directed to improving chemical vapor deposition (CVD) boron uniformity overcoming loading effects. The loading effects of boron layers on silicon within a window may be reduced or eliminated by depositing an adhesion layer on a dielectric layer before depositing the boron layer. The adhesion layer may reduce or eliminate lateral diffusion of

boron species into the window by being deposited on the adhesion layer. The approach using the adhesion layer may enable forming the boron layer at the nanometer scale within windows which are at the tens of millimeters scale and below. The boron layer and the silicon layer may form a detector which may be used in scanning electron microscopes and the like.

[0029] U.S. Pat. No. 6,555,166B2, titled “Method for reducing the microloading effect in a chemical vapor deposition reactor”; U.S. Pat. No. 7,586,108B2, titled “Radiation detector, method of manufacturing a radiation detector and lithographic apparatus comprising a radiation detector”; U.S. Pat. No. 8,450,820B2, titled “Radiation detector”; U.S. Pat. No. 9,818,887B2, titled “Back-illuminated sensor with boron layer”; U.S. Pat. No. 10,197,501B2, titled “Electron-bombarded charge-coupled device and inspection systems using EBCCD detectors”; U.S. Pat. No. 10,199,197B2, titled “Photocathode including silicon substrate with boron layer”; U.S. Pat. No. 10,269,842B2, titled “Anti-reflection layer for back-illuminated sensor”; U.S. Pat. No. 10,466,212B2, titled “Scanning electron microscope and methods of inspecting and reviewing samples”; U.S. Pat. No. 11,088,190B2, titled “Optical semiconductor device”; U.S. Pat. No. 11,114,489B2, titled “Back-illuminated sensor and a method of manufacturing a sensor”; U.S. Pat. No. 11,239,048B2, titled “Arrayed column detector”; U.S. Pat. No. 11,699,607B2, titled “Segmented multi-channel, backside illuminated, solid state detector with a through-hole for detecting secondary and backscattered electrons”; U.S. Pat. No. 11,848,350B2, titled “Back-illuminated sensor and a method of manufacturing a sensor using a silicon on insulator wafer”; U.S. Patent Publication Number US20200212246A1, titled “Semiconductor detector and method of fabricating same”; U.S. Patent Publication Number US20230230800A1, titled “Sensor module for scanning electron microscopy applications”; U.S. Patent Publication Number US20240243143A1, titled “Boron-Coated Back-Illuminated Image Sensor With Fluoride-Based Anti-Reflection Coating”; U.S. Patent Publication Number US20240250110A1, titled “Image sensors with a tunable floating diffusion structure”; U.S. Patent Publication Number US20240313032A1, titled “Back-Illuminated Sensor With Boron Layer Deposited Using Plasma Atomic Layer Deposition”; are each incorporated herein by reference in the entirety.

[0030] FIGS. **1A-1B** depicts a detector **100**, in accordance with one or more embodiments of the present disclosure. The detector **100** may be configured to detect electrons **101** and/or photons **103**. The photons **103** may include vacuum-ultraviolet (VUV) light and/or soft X-rays where the detector **100** is configured as a photon detector. The detector **100** may include a silicon layer **102**, a dielectric layer **104**, an adhesion layer **106**, boron layers **108**, an epitaxial layer **110**, an anode **112**, a cathode **114**, and/or a capping layer **116**.

[0031] The silicon layer **102** may be monocrystalline silicon (i.e., monosilicon). The silicon layer **102** may be an n-type semiconductor. The silicon layer **102** may be silicon doped with donor atoms to be the n-type semiconductor. The silicon layer **102** may be doped with any suitable donor atom, such as, but not limited to, phosphorous, arsenic, or the like. The silicon layer **102** may include a select doping concentration of the donor atoms. The silicon layer **102** may be heavily doped with the donor atoms. The silicon layer **102** may be an (n⁺)-type semiconductor.

[0032] The silicon layer **102** may be a substrate over which the dielectric layer **104**, the adhesion layer **106**, the boron layers **108**, the epitaxial layer **110**, the anode **112**, the cathode **114**, and/or the capping layer **116** are deposited.

[0033] The epitaxial layer **110** may be an n-type semiconductor. The epitaxial layer **110** may be doped with a select concentration of the donor atoms. The epitaxial layer **110** may be lightly doped. The epitaxial layer **110** may be a (n⁻)-type semiconductor. The doping concentration of the epitaxial layer **110** may be less than the doping concentration of the silicon layer **102**. For example, the doping concentration of the epitaxial layer **110** may be an order of magnitude smaller than the silicon layer **102**.

[0034] The epitaxial layer **110** may be deposited on the silicon layer **102**. The epitaxial layer **110** may be disposed between and couple the silicon layer **102** and either or both the dielectric layer

104 and/or the lower boron layer **108b**.

[0035] The dielectric layer **104** may be a dielectric material. The dielectric layer **104** may be any silicon-based dielectric material. For example, the dielectric layer **104** may be silicon oxide (e.g., silicon dioxide), silicon nitride, or the like. In embodiments, the dielectric layer **104** is silicon oxide. The silicon oxide may be amorphous.

[0036] The dielectric layer **104** may be deposited on the epitaxial layer **110**. The dielectric layer **104** may be disposed between and couple the epitaxial layer **110** and the adhesion layer **106**.

[0037] The adhesion layer **106** may be material that may react with boron to consume diborane radicals. The adhesion layer **106** may be a material on which diborane gas may be decomposed and deposited at the same or similar rates as on the epitaxial layer **110**. The adhesion layer **106** may be a Group IV material, such as carbon, silicon, germanium, tin, lead, or a combination thereof. In embodiments, the adhesion layer **106** may be a silicon-based Group IV material. The silicon-based Group IV material includes silicon, may include another Group IV material, and may not include any other materials which are not Group IV materials. For example, the adhesion layer **106** may be polycrystalline silicon (i.e., polysilicon), silicon germanium (SiGe), silicon carbide (SiC), silicon tin (SiSn), silicon lead (SiPb), or the like. In embodiments, the adhesion layer **106** is polycrystalline silicon. The polycrystalline silicon may be amorphous. It is contemplated that polycrystalline silicon may be advantageous due boron being deposited on the polycrystalline silicon and monocrystalline silicon at the same rate.

[0038] The adhesion layer **106** may be deposited on the dielectric layer **104**. The adhesion layer **106** may be disposed between and couple the dielectric layer **104** and the boron layer **108**.

[0039] The adhesion layer **106** may include a select thickness. The thickness of the adhesion layer **106** may be on the order of hundreds of nanometers or micrometers (e.g., between 100 nm and 9 μ m). For example, the thickness of the adhesion layer **106** may be on the order of hundreds of nanometers (e.g., between 100 nm and 999 nm). The thickness of the adhesion layer **106** may not control the deposition rate of the boron layer **108** on the adhesion layer **106**. Thus, the adhesion layer **106** may be made as thin as possible with no effects on the performance of the detector **100**. For example, the adhesion layer **106** was experimentally verified to enable the deposition of the boron layer **108** on the adhesion layer **106** with thicknesses as small as 100 nm and 200 nm.

[0040] A window **107** may be defined by the dielectric layer **104** and the adhesion layer **106**. Sides of the dielectric layer **104** and the adhesion layer **106** may define the window **107**. The window **107** may be defined through the dielectric layer **104** and the adhesion layer **106** to the epitaxial layer **110**. The dielectric layer **104** and the adhesion layer **106** may be etched to define the window **107**. The window **107** may or may not be defined through at least a portion of the epitaxial layer **110**. For example, the portion of the epitaxial layer **110** may or may not be etched with the dielectric layer **104** and the adhesion layer **106** during etching.

[0041] The window **107** may include a width. The width of the window may be a distance between the sides of the dielectric layer **104** and/or the adhesion layer **106** defining the window **107**. The width of the window **107** may be on the order of hundreds of micrometers, millimeters, tens of millimeters, or hundreds of millimeters (e.g., between 100 μ m and 999 mm). In embodiments, the width of the window **107** may be on the order of hundreds of micrometers, millimeters, or tens of millimeters (e.g., between 100 μ m and 99 mm). In embodiments, the width of the window **107** may be on the order of hundreds of micrometers or millimeters (e.g., between 100 μ m and 9 mm). The window **107** has been experimentally verified for widths as small as 600 μ m. The adhesion layer **106** may enable the width of the window **107** to be smaller than hundreds of millimeters while reducing the thickness of the boron layer **108**, as will be described further herein. Reducing the width of the window **107** may cause the detector **100** to be smaller than existing detectors and/or provide high bandwidth detection.

[0042] The width of the window **107** may be constant or vary along a height of the window **107**. For example, the sides of the dielectric layer **104** and/or the adhesion layer **106** may be flat such

that the width of the window **107** may be constant (as depicted). By way of another example, the sides of the dielectric layer **104** and/or the adhesion layer **106** may be beveled such that the width of the window **107** may decrease from the top of the adhesion layer **106** to the epitaxial layer **110**. [0043] The boron layers **108** may be amorphous. The boron layers **108** may also be a p-type semiconductor. The boron layers **108** may be heavily doped. For example, the boron layers **108** may be a (p+)-type semiconductor.

[0044] The boron layers **108** may be high-purity boron. The boron layers **108** may include a high concentration of boron. For example, the boron layers **108** may include a boron concentration of at least **80** percent, with inter-diffused silicon and oxygen predominantly making up the remaining percentage. In embodiments, the boron layers **108** may be a pure boron layer (e.g., pure-B). For example, the boron layers **108** may include a boron concentration of at least 95 percent. For instance, the boron layers **108** may include a boron concentration of at least 99 percent. The purity of the lower boron layer **108b** may be important for the sensitivity and lifetime of the detector **100**. If the boron layers **108** includes a native oxide, the native oxide may be affected by the electrons **101** and/or photons **103** and may cause a degradation of performance. Therefore, increasing the boron concentration may be desirable.

[0045] The boron layers **108** may be deposited on the epitaxial layer **110** within the window **107** and/or on the adhesion layer **106**. The boron layers **108** may include an upper boron layer **108a** and/or a lower boron layer **108b**. The upper boron layer **108a** may be deposited on the adhesion layer **106**. The lower boron layer **108b** may be deposited on the epitaxial layer **110** within the window **107**. A portion of the upper boron layer **108a** deposited on the adhesion layer **106** may or may not overhang and be deposited on the sides of the adhesion layer **106** defining the window **107**.

[0046] The boron layers **108** may not be deposited on the dielectric layer **104**. The upper boron layer **108a** may be separated from the lower boron layer **108b** by the dielectric layer **104**. Thus, the dielectric layer **104** may insulate the upper boron layer **108a** from the lower boron layer **108b**. Insulating the upper boron layer **108a** from the lower boron layer **108b** may be advantageous to maintain functionality of the detector **100** (e.g., to allow holes to move to the lower boron layer **108b** as will be described further herein).

[0047] The lower boron layer **108b** may include a select thickness. The thickness of the lower boron layer **108b** may depend on the intended application for the detector **100**. The thickness of the lower boron layer **108b** may be on the order of nanometers or tens of nanometers (e.g., between 1 nm and 99 nm). The thickness of the lower boron layer **108b** may be between a minimum thickness tolerance and a maximum thickness tolerance. For example, the thickness of the lower boron layer **108b** may be between 2 nm and 20 nm. For instance, the thickness of the lower boron layer **108b** may be between 2 nm and 10 nm. In embodiments, the thickness of the lower boron layer **108b** may be between 2 nm and 5 nm. In embodiments, the thickness of the lower boron layer **108b** may be between 2 nm and 3 nm.

[0048] The lower boron layer **108b** may be a pinhole-free film. Providing the thickness of the lower boron layer **108b** above the minimum thickness tolerance may cause the lower boron layer **108b** to be the pinhole-free film. A thickness of 2 nm or greater may ensure the lower boron layer **108b** is pinhole-free. The pinhole-free film may prevent oxygen from passing through the lower boron layer **108b** to the epitaxial layer **110** and oxidizing, thereby reducing the sensitivity and/or lifetime of the detector **100**.

[0049] The electrons **101** and/or the photons **103** may penetrate the lower boron layer **108b**. Providing the thickness of the lower boron layer **108b** below the maximum thickness tolerance may allow the penetration through the lower boron layer **108b** by the electrons **101** and/or the photons **103**. The maximum thickness tolerance of the lower boron layer **108b** may be less than the penetration depth of the electrons **101** and/or the photons **103**. Reducing the thickness of the lower boron layer **108b** below the maximum thickness tolerance may also be beneficial to improve the

sensitivity and/or responsivity of the detector **100**.

[0050] The upper boron layer **108a** may also include a select thickness. The thickness of the lower boron layer **108b** may be on the order of nanometers or tens of nanometers. The upper boron layer **108a** may include a same or similar thickness as the lower boron layer **108b**. For example, the deposition rate at which boron is deposited on the epitaxial layer **110** and the adhesion layer **106** may be the same or similar such that the upper boron layer **108a** may include the same or similar thickness as the lower boron layer **108b**. Thus, the thicknesses of the upper boron layer **108a** and the lower boron layer **108b** may be on the on the order of nanometers or tens of nanometers, and may be further understood with the examples and illustrations provided above in regards to the lower boron layer **108b**.

[0051] The thickness of the lower boron layer **108b** may vary between the minimum thickness and the maximum thickness along the width of the window **107**. For example, the lower boron layer **108b** may experience local loading effects. The local loading effects may be pattern dependent. The local loading effects may cause the thickness of the lower boron layer **108b** to vary along the width of the window **107**. For example, the local loading effects may cause the maximum thickness of the lower boron layer **108b** to be at the interface between the lower boron layer **108b** and the minimum thickness of the lower boron layer **108b** to be around the midpoint of the window **107**. The impact of the local loading effects may increase as the size of the window **107** decreases.

[0052] The adhesion layer **106** and the upper boron layer **108a** may reduce or eliminate the effect of local loading and/or improve the uniformity in thickness of the lower boron layer **108b** along the width of the lower boron layer **108b**. The adhesion layer **106** may reduce or eliminate the effect of local loading by reacting with depositing the boron in diborane gas on the adhesion layer **106**, thereby reducing and/or eliminating the lateral diffusion of the boron into the window **107**. With the adhesion layer **106**, the lower boron layer **108b** is deposited much more homogeneously than without the adhesion layer **106**. For example, the adhesion layer **106** and the upper boron layer **108a** may enable the detector **100** to include the window **107** with a width on the order of several hundreds of micrometers, millimeters, tens of millimeters, or hundreds of millimeters while maintaining the thickness of the lower boron layer **108b** within the minimum thickness tolerance and the maximum thickness tolerance along the width of the window **107**. Thus, the adhesion layer **106** and the upper boron layer **108a** may decorrelate the local loading effects to the width of the window **107**.

[0053] The adhesion layer **106** and the upper boron layer **108a** may be conductive and act as an electron shield for the dielectric layer **104**. For example, the upper boron layer **108a** may dope the adhesion layer **106** to be conductive. The dielectric layer **104** may negatively charge by trapping electrons in response to receiving the electrons. The adhesion layer **106** and/or the upper boron layer **108a** may conduct the negative charge away from the dielectric layer **104**. The adhesion layer **106** and the upper boron layer **108a** may eliminate or greatly reduce the negative charging of the dielectric layer **104** due to electron bombardment or exposure by conducting the electrons away from the dielectric layer **104**. Thus, the adhesion layer **106** and the upper boron layer **108a** may act as the electron shield for the dielectric layer **104**.

[0054] The anode **112** and cathode **114** may be coupled to the lower boron layer **108b** and the silicon layer **102**, respectively. The anode **112** and/or the cathode **114** may be deposited on a same side of the silicon layer **102** (not depicted). For example, the dielectric layer **104**, the anode **112**, and/or the cathode **114** may each be deposited on a same side of the silicon layer **102**. The anode **112** and/or the cathode **114** may be deposited on the silicon layer **102** as interconnect pads (not depicted) or the like. The anode **112** and/or the cathode **114** may be a conductive metal. For example, the anode **112** and/or the cathode **114** may be aluminum, copper, or the like.

[0055] The lower boron layer **108b**, the epitaxial layer **110**, and the silicon layer **102** may form a p-n diode. For example, the p-n diode may be a (p+)(n-)(n+) diode where the epitaxial layer **110** is the (n-)-type semiconductor. The p-n diode may be the sensing element of the detector **100** by

which the detector **100** is configured to detect the electrons **101** and/or the photons **103**. A depletion region (not depicted) of the p-n diode may be disposed within the epitaxial layer **110**. The electrons **101** and/or the photons **103** may penetrate through the lower boron layer **108b** to the epitaxial layer **110** (e.g., to the depletion region). The epitaxial layer **110** may generate electron-hole pairs in response to absorbing the electrons **101** and/or the photons **103**. The number of electron-hole pairs generated may be proportional to the energy of the electrons **101** and/or the photons **103**. The electron-hole pairs may move from the epitaxial layer **110** by an electric field caused by the depletion region. The holes may move toward the anode **112** through the lower boron layer **108b** and the electrons may move toward the cathode **114** through the silicon layer **102**. The flow of electron-hole pairs may induce a current between the lower boron layer **108b** and the silicon layer **102**. The current may be collected and readout from the anode **112** and/or cathode **114** as a signal for detecting the electrons **101** and/or the photons **103**. Thus, the detector **100** may be configured to detect the electrons **101** and/or the photons **103**.

[0056] The capping layer **116** may be deposited on the lower boron layer **108b**. The capping layer **116** may protect and/or prevent degradation of the lower boron layer **108b**. The improved uniformity in thickness of the lower boron layer **108b** may also improve a uniformity in thickness of the capping layer **116**.

[0057] The capping layer **116** may also be deposited on the dielectric layer **104** and/or the upper boron layer **108a** (not depicted). For example, the capping layer **116** may be conformal to the dielectric layer **104**, the upper boron layer **108a**, and/or the window **107**.

[0058] The capping layer **116** may be compatible with detecting the electrons **101** and/or the photons **103**. The electrons **101** and/or the photons **103** may penetrate through the capping layer **116**. Where the detector **100** is configured to detect the electrons **101**, the capping layer **116** may be a metal, such as aluminum oxide, ruthenium, tungsten, or molybdenum. The metal may also be a spectral filter to allow the penetration of the electrons **101** while preventing the penetration of the photons **103**. Where the detector **100** is configured to detect the photons **103**, the capping layer **116** may be an anti-reflective (AR) coating. The anti-reflection coating may be a VUV anti-reflection coating. The anti-reflection coating may include one or more materials, such as silicon dioxide, silicon nitride, aluminum oxide, magnesium fluoride, lithium fluoride, hafnium dioxide, or the like.

[0059] FIG. 2 depicts a flow diagram of a method **200**, in accordance with one or more embodiments of the present disclosure. The method **200** is a method of forming the detector **100**. The embodiments and enabling technologies described previously herein in the context of the detector **100** should be interpreted to extend to method **200**. It is further noted, however, that the method is not limited to the architecture of the detector **100**. The steps of the method **200** may be further understood with reference to FIGS. 3A-3G.

[0060] In a step **205**, an epitaxial layer may be deposited on a silicon layer. For example, the epitaxial layer **110** may be deposited on the silicon layer **102**. The epitaxial layer **110** may be deposited using any suitable technique.

[0061] In a step **210**, a dielectric layer may be deposited on the epitaxial layer. For example, the dielectric layer **104** may be deposited on the epitaxial layer **110**. The dielectric layer **104** may be deposited using any suitable technique. For example, the dielectric layer **104** may be silicon oxide which may be deposited by thermal growth at a temperature, such as approximately 900 to 1000° C. The silicon oxide may be stable when grown to a thickness of about 1.5 nm to 2 nm. By way of another example, the dielectric layer **104** may be silicon nitride which may be deposited using plasma-enhanced chemical-vapor deposition (CVD).

[0062] In a step **220**, an adhesion layer may be deposited on the dielectric layer. For example, the adhesion layer **106** may be deposited on the dielectric layer **104**. The dielectric layer **104** may be deposited using any suitable technique. For example, the dielectric layer **104** may be polycrystalline silicon which may be deposited by chemical vapor deposition.

[0063] The deposition of the adhesion layer **106** on the dielectric layer **104**, the deposition of the

dielectric layer **104** on the epitaxial layer **110**, and/or the deposition of the epitaxial layer **110** on the silicon layer **102** may be blanket-type depositions where the adhesion layer **106**, the dielectric layer **104**, and/or the epitaxial layer **110** cover the width of the silicon layer **102**, without any pre-patterning.

[0064] In a step **230**, a window may be etched through the adhesion layer and the dielectric layer to the epitaxial layer. For example, the window **107** may be etched through the adhesion layer **106** and the dielectric layer **104** to the epitaxial layer **110**. Both the adhesion layer **106** and the dielectric layer **104** may be patterned by a same mask.

[0065] The window **107** may be etched using any suitable technique. For example, the window **107** may be etched by wet etching or dry etching. The etching can be performed by wet etching methods using alkaline solutions such as potassium hydroxide (KOH), tetramethylammonium hydroxide (TMAH), or the like. The adhesion layer **106** may be first etched with the dielectric layer **104** acting as a natural etch stop. The dielectric layer **104** may then be removed using hydrofluoric acid and/or buffered oxide etch with the epitaxial layer **110** acting as a natural etch stop.

Alternatively, the adhesion layer **106** and the dielectric layer **104** may be removed using dry etching methods such as reactive ion etching (RIE) and/or inductively coupled plasma reactive ion etching (ICPRIE).

[0066] In a step **240**, boron layers may be deposited on the epitaxial layer and the adhesion layer. For example, the upper boron layer **108a** may be deposited on the adhesion layer **106** and the lower boron layer **108b** may be deposited on the epitaxial layer **110**. The epitaxial layer **110** may include an oxide-free surface on which the lower boron layer **108b** is deposited.

[0067] The boron layers **108** may be deposited using any suitable technique. For example, the boron layers **108** may be deposited using diborane (B_2H_6) in an atmospheric-/reduced-pressure chemical vapor deposition (AP/PCVD) reactor. A main stream gas **302** may flow over the detector **100**. The main stream gas **302** may include a gas source and a carrier gas. Diborane and hydrogen may be used as the gas source and the carrier gas, respectively. The main stream gas **302** may be a mixture of diborane and hydrogen gases at a temperature to create the high-purity amorphous boron layers. The temperature may be higher than 500°C . For example, the temperature may be between 600 and 900°C . For instance, the temperature may be between 700 and 800°C . The boron layers **108** may also be deposited at lower temperatures, although the thickness may be less uniform. A boundary layer **304** of the carrier gas (e.g., the hydrogen gas) may be formed between the main stream gas **302** and the detector **100**. Diborane species **306** may decompose from the diborane gas and may diffuse through the boundary layer **304**. The diborane species **306** may vertically diffuse **308** and be deposited as the upper boron layers **108a** on the adhesion layer **106** and as the lower boron layers **108b** on the epitaxial layer **110**. The adhesion layer **106** and the epitaxial layer **110** may react with the diborane species **306** to deposit the upper boron layer **108a** and the lower boron layer **108b**, respectively. The dielectric layer **104** may not react with the diborane species **306**, such that boron layers **108** are not deposited on the dielectric layer **104**.

[0068] The diborane species **306** may or may not laterally diffuse **310** along the adhesion layer **106** into the window **107**. The adhesion layer **106** may suppress the lateral diffusion. For example, the deposition of the upper boron layers **108a** on the adhesion layer **106** may prevent the lateral diffusion into the window **107**. By eliminating the lateral diffusion, there is no excessive radicals around the edge of the window **107**. The uniformity in thickness of the lower boron layer **108b** may then be achieved for any size of the window **107**.

[0069] In an optional step **250**, a capping layer may be deposited on the lower boron layer. For example, the capping layer **116** may be deposited on the lower boron layer **108b**. The capping layer **116** may also be deposited on the upper boron layer **108a** and/or the sides of the window **107**. The capping layer **116** may be deposited using any suitable technique. For example, the capping layer **116** may be deposited using atomic layer deposition (ALD). The atomic layer deposition may

enable deposition of the capping layer **116** with very precise (e.g., single monolayer) control of the thickness.

[0070] In an optional step **255**, the upper boron layer and the adhesion layer may be removed. For example, the upper boron layer **108a** and the adhesion layer **106** may be removed. The upper boron layer **108a** and the adhesion layer **106** may be removed using any suitable technique. For example, the upper boron layer **108a** and the adhesion layer **106** may be removed by polishing and/or etching. With the lower boron layer **108b** being protected by resist or other masking material, the upper boron layer **108a** and the adhesion layer **106** may be removed using chlorine-containing chemistry by reactive-ion etching (RIE), or the like. The upper boron layer **108a** and the adhesion layer **106** may be etched using etching chemistries with selectivity to polysilicon-over-oxide. For instance, the window **107** may be dry etched using chlorine trifluoride gas or the like.

[0071] The upper boron layer **108a** and the adhesion layer **106** may be removed to expose the dielectric layer **104**. Exposing the dielectric layer **104** by removing the upper boron layer **108a** and the adhesion layer **106** may ensure the compatibility of the detector **100** with one or more further processing steps. It is further contemplated that the detector **100** may be compatible with the further processing steps without removing the upper boron layer **108a** and the adhesion layer **106**.

[0072] In a step **260**, further processing may be performed to complete the detector **100**. Additional steps may be performed on the front side that finish the detector **100**. The further processing steps may include forming interconnects (e.g., including the anode **112** and the cathode **114**). The interconnects may be formed by Al, Cu, or another metal. The interconnects may also be covered by a passivation layer to protect the interconnects. The subsequent processing steps may include packaging the detector **100**. The package may include flip-chip bonding or wire bonding of a chip to a substrate. The package may include a window that transmits wavelengths of interest, or may comprise a flange or seal for interface to a vacuum seal. In electron-bombarded image sensor embodiments, the package may include other components such as a photocathode as well as a sealed, evacuated tube. The resulting structure may be packed in a suitable package. The packing step may comprise flip-chip bonding or wire bonding of the device to the substrate. The package may include a window that transmits wavelengths of interest, or may include a flange or seal for interface to a vacuum seal.

[0073] FIGS. **4A-4C** depicts the detector **100**, in accordance with one or more embodiments of the present disclosure. The detector **100** may be configured to detect secondary electrons (SE) and/or backscatter electrons (BSE).

[0074] The detector **100** may be annular. The silicon layer **102**, the dielectric layer **104**, the adhesion layer **106**, and/or the upper boron layer **108a** may define the annular shape with the lower boron layer **108b** disposed radially therebetween.

[0075] The detector **100** may define an aperture **402**. The aperture **402** may be concentric to the annular shape. The aperture **402** may be defined through the detector **100**. For example, the aperture **402** may be defined through the silicon layer **102**, the dielectric layer **104**, the adhesion layer **106**, the upper boron layer **108a**, and/or the epitaxial layer **110**. The aperture **402** may permit an electron beam (e.g., electron beam **506**) to pass through the detector **100**. The lower boron layer **108b** may be concentric too and surround the aperture **402** for receiving the electrons **101** in response to the electron beam. The detector **100** may be designed for backside illumination, meaning the electron beam may pass through the aperture **402**, hit a sample (e.g., sample **528**), and cause secondary electrons and/or backscattered electrons to reflect toward the detector **100** and hit the lower boron layer **108b**.

[0076] The detector **100** may also be segmented into multiple channels. The channels may be configured to independently detect the electrons **101** and/or the photons **103**. The channels may provide increased spatial resolution when detecting the electrons **101**. The multiple channels may also allow the detector **100** to simultaneously detect secondary electrons and backscattered electrons. Each of the channels may be configured to have equal area for equal capacitance. The

equal capacitance may allow the channels to have equal or very similar responses as the other channels to optimize the electronic readout. The channels may each include the lower boron layer **108b** and the epitaxial layer **110** which are segmented by the dielectric layer **104**, the adhesion layer **106**, and the upper boron layer **108a**. The detector **100** may include any number of the channels. For example, FIG. **4C** depicts the detector **100** segmented into five channels. The channels may include a center channel (CH0) surround by four quadrants of channels (CH1, CH2, CH3, CH4). The aperture **402** may be through a center of the center channel (CH0).

[0077] FIG. **5** depicts a scanning electron microscope **500** (SEM), in accordance with one or more embodiments of the present disclosure. The scanning electron microscope **500** may be an inspection or review system configured to inspect or review a sample **528**. The scanning electron microscope **500** may review and/or inspect the sample **528** for defects and to reveal the material composition of the sample **528** and/or defects. The scanning electron microscope **500** may include an electron source **502**, an electron optical system **511**, a stage **530**, and/or a controller **540**.

[0078] The electron source **502** may be a source of electrons. The electron source **502** may include any electron source suitable for generating an electron beam **506**. The electron source **502** may also be referred to as an electron gun. The electron beam **506** may also be referred to as a primary electron beam. The electron beam **506** may include a beam energy and a beam current.

[0079] The electron source **502** may include electron emitters **501**, extractors **503**, a magnetic lens (not depicted), and the like. The electron source **502** may include any number of the electron emitters **501**. For example, the electron source **502** may include a single of the electron emitters **501**. By way of another example, the electron source **502** may include multiple of the electron emitters **501**. The electron emitters **501** may be a cathode. For example, the cathode may include a thermal field-emitting a Schottky cathode, a cold-field emitter, a single-crystal tungsten cathode, a LaB6 cathode, or the like. The extractors **503** may be electrodes. The extractors **503** may be arranged in the path of the electron beam **506**.

[0080] The electron optical system **511** may include a set of electron optics arranged in an electron-optical column. The electron optical system **511** may include one or more focusing optics and/or deflecting optics. The focusing optics may focus the electron beam **506** onto the sample **528**. The electron optical system **511** may include any focusing optics. For example, the one or more focusing optics may include, but are not limited to, condenser lens **507** and objective lens **510**.

[0081] The condenser lens **507** may de-magnify the electron beam **506** to create a small spot on the sample **528**. Spot sizes of about one or a few nm may generate high-resolution images for review of the sample **528**. Inspection of the sample **528** may use larger spot sizes to scan the sample **528** more quickly. The electron optical system **511** may include any number of the condenser lens **507**. A single of the condenser lens **507** may suffice when the spot size is of order of one-hundred nanometers or larger, but two or more of the condenser lens **507** may be needed for spot sizes of tens of nm or smaller. The condenser lens **507** may include a magnetic lens, an electrostatic lens, both a magnetic lens and an electrostatic lens, or the like.

[0082] The objective lens **510** may be a final lens within the electron optical system **511** before the electron beam **506** reaches the sample **528**. The objective lens **510** may focus the electron beam **506** to a small spot on the sample **528**. The objective lens **510** may include a magnetic lens, an electrostatic lens, a combination of a magnetic lens and an electrostatic lens, or the like. The objective lens **510** may be an immersion lens. To achieve a small spot size at the sample **528**, the objective lens **510** may be placed close to the sample **528**, so that the sample **528** is immersed in the magnetic field of the objective lens **510**. Immersing the sample in the magnetic field may reduce aberrations in the electron spot on the sample **528**.

[0083] The deflection optics may be configured to scan the electron beam **506** over the sample **528**. The electron optical system **511** may include any deflection optics. The one or more deflection optics may include, but are not limited to, one or more deflectors (e.g., scanning coils). For example, the electron optical system **511** may include deflectors **505** and deflectors **509**. The

deflectors **505** may scan the electron beam **506** over an area of the sample **528**. The deflectors **505** may be placed on either side of condenser lens **507** as shown, or within the condenser lens **507** (not shown), or after the condenser lens **507**. The deflectors **505** may include electrostatic deflectors or a combination of magnetic and electrostatic deflectors. The deflectors **509** may work in combination with deflectors **505** to scan the electron beam **506** over an area of the sample **528**.

[0084] The electron beam **506** may be focused and deflected by the deflectors **505**, the condenser lens **507**, the deflectors **509**, and the objective lens **510** onto the sample **528** positioned on the stage **530**. The electron source **502** and electron optical system **511** may be arranged in a single-beam configuration or in a multi-beam configuration. For example, the electron source **502** may generate a single of the electron beam **506** that is deflected and focused onto the sample **528**. By way of another example, the electron source **502** may also generate multiple of the electron beams **506** that are deflected and focused onto the sample **528**. As depicted, the electron source **502** and electron optical system **511** are depicted in the single-beam configuration, although this is not intended to be limiting.

[0085] The sample **528** may include sample types typically reviewed by the scanning electron microscope **500**. For example, the sample **528** may include an un-patterned semiconductor wafer, a patterned semiconductor wafer, a reticle, a photomask, or the like.

[0086] The stage **530** may support and position the sample **528**. The sample **528** may be placed on the stage **530**. The stage **530** may facilitate movement of different regions of sample **528** underneath the electron optical system **511**. The stage **530** may include an X-Y stage or an R-0 stage. The stage **530** may adjust the height of sample **528** during inspection to maintain focus.

[0087] The electron optical system **511** may also include a focus sensor (not shown). The focus sensor may be an optical sensor. The focus sensor may also be referred to as a height sensor. The focus sensor may be mounted on or proximate to objective lens **510**. The focus sensor may provide a signal to adjust the height of sample **528** or to adjust the focus of the objective lens **510**.

[0088] The electrons **101** may be emitted from an area of the sample **528** when the electron beam **506** is scanned by the electron optical system **511** over the area. The electrons **101** may emanate from the sample **528** in response to the electron beam **506**. The electrons **101** may be secondary and/or backscattered electrons. Secondary electrons and back-scattered electrons may be emitted from an area of the sample **528** when the electron beam **506** is scanned by the electron optics across the area.

[0089] The detector **100** may not block the electron beam **506** on the way to the sample **528**. The detector **100** may be positioned at one or more selected locations in view of the sample **528**. The detector **100** may be configured to collect the electrons **101** emanated from the sample **528** in response to the electron beam **506**. The electrons **101** may be detected by the detector **100** that is in direct line-of sight to the region of the sample **528** being exposed to the electron beam **506**. The detector **100** may be adjacent to the sample **528**. For example, the detector **100** may be placed between the objective lens **510** and the sample **528**. The gap between the sample **528** and the objective lens **510** may be small, such as about 2 mm or less, and clearance may be needed, for example, for a focus or height sensor, and so it may not be practical to place the detector **100** between the objective lens **510** and the sample **528**.

[0090] The scanning electron microscope **500** may include any number of the detectors **100**. The detectors **100** may also be multi-column electron detectors. For example, the scanning electron microscope **500** may include multiple electron beam columns with the multi-column electron detectors. The multiple electron beam columns may enable parallel inspection of the sample **528**.

[0091] The detector **100** may include an application specific integrated circuit. The application specific integrated circuit may be used to digitize and preprocess the signals collected by the detector **100**. The detector **100** may be configured to generate image data signal ID in response to detecting the electrons **101**. The image data signal ID may be transferred to controller **540** from the detector **100**. The controller **540** may utilize the detector **100** to generate images of the scanned

area of the sample **528**. For example, the controller **540** may utilize the X-ray image data signal ID to generate an image of the associated scanned sample area, whereby visual inspection of a defect including an unknown material is facilitated.

[0092] The controller **540** may also be configured to perform additional functions, such as determining the presence of a defect and/or the type of the defect based on the image data signals ID. For example, the controller **540** may use the image data signals ID to determine the presence of the defect in the scanned area. The defect may include the presence of material (such as a particle) that is not supposed to be present in the scanned area, the absence of material that is supposed to be in the scanned area (such as may happen with an over-etched condition), or a malformed pattern. The controller **540** may also determine the defect type and/or material type of the defect.

[0093] FIG. **6** depicts a detector array **600**, in accordance with one or more embodiments of the present disclosure. The detector array **600** may include an array of the detectors **100**. The detectors **100** may be arranged in a two-dimensional array with a select lattice configuration. For example, the detectors **100** may be arranged in a hexagonal lattice (as depicted), a square lattice, or the like. The detectors **100** may be arranged with a select pitch between adjacent of the detectors **100**. For example, the pitch between the detectors **100** may be on the order of micrometers, tens of micrometers, or hundreds of micrometers (e.g., between 1 and 999 micrometers). The silicon layer **102** and/or the epitaxial layer **110** may be monolithic across the detector array **600**. In this regard, the silicon layer **102** and/or the epitaxial layer **110** may be a single piece which is not formed of constituent pieces which are patterned. The silicon layer **102** may be a wafer or reticle. The dielectric layer **104**, the adhesion layer **106**, and/or the boron layers **108** may be patterned across the detector array **600** to define the array of the detectors **100**.

[0094] The detectors **100** in the detector array **600** may be configured to detect secondary electrons (SE) and/or backscatter electrons (BSE) and/or may be segmented into the channels.

[0095] The detectors **100** in the detector array **600** may be formed by etching the windows **107** for each of the detectors **100** when etching the windows **107** to the epitaxial layer **110**.

[0096] The scanning electron microscope **500** may include the detector array **600**, where each of the detectors **100** may be configured to generate the image data signal ID in response to detecting the electrons **101**. The array of the detectors **100** may be advantageous to achieve low crosstalk between different beam channels of the scanning electron microscope **500**. Reducing the crosstalk may be beneficial to decrease a signal-to-noise ratio (SNR) of the scanning electron microscope **500**.

[0097] Referring generally again to the figures.

[0098] Although the epitaxial layer **110** is described as an n-type semiconductor and the p-n diode is described as a being a (p+)(n-)(n+) diode, this is not intended as a limitation of the present disclosure. It is contemplated that the benefits of the adhesion layer **106** may apply to various types of p-n diodes. For example, the epitaxial layer **110** may be an intrinsic layer and the p-n diode may be a PIN diode. By way of another example, the epitaxial layer **110** may be an lightly-doped p-type layer (e.g., a (p-)-type semiconductor) and the p-n diode may be a (p+)(p-)(n+) diode, where the doping concentration of the lightly-doped p-type layer is less than the lower boron layer **108b**. The intrinsic layer and/or the lightly-doped p-type layer may be grown by forming a thin-boron doped layer at low temperatures (e.g., less than or equal to 450° C.) before depositing the boron layers **108**.

[0099] It is further contemplated that each of the embodiments of the method described above may include any other step(s) of any other method(s) described herein. In addition, each of the embodiments of the method described above may be performed by any of the systems described herein.

[0100] In embodiments, a controller may include one or more controllers housed in a common housing or within multiple housings. In this way, any controller or combination of controllers may be separately packaged as a module suitable for integration into a system. Further, the controllers

may analyze data received from detectors and feed the data to additional components within the system or external to the system. The controller may also include one or more processors and/or a memory medium.

[0101] The one or more processors may include any processor or processing element known in the art. For the purposes of the present disclosure, the term “processor” or “processing element” may be broadly defined to encompass any device having one or more processing or logic elements (e.g., one or more micro-processor devices, one or more application specific integrated circuit (ASIC) devices, one or more field programmable gate arrays (FPGAs), or one or more digital signal processors (DSPs)). In this sense, the one or more processors may include any device configured to execute algorithms and/or instructions (e.g., program instructions stored in memory). In one embodiment, the one or more processors may be embodied as a desktop computer, mainframe computer system, workstation, image computer, parallel processor, networked computer, or any other computer system configured to execute a program. Moreover, different subsystems of the system may include a processor or logic elements suitable for carrying out at least a portion of the steps described in the present disclosure. Therefore, the above description should not be interpreted as a limitation on the embodiments of the present disclosure but merely as an illustration. Further, the steps described throughout the present disclosure may be carried out by a single controller or, alternatively, multiple controllers.

[0102] The memory medium may include any storage medium known in the art suitable for storing program instructions executable by the associated one or more processors. For example, the memory medium may include a non-transitory memory medium. By way of another example, the memory medium may include, but is not limited to, a read-only memory (ROM), a random-access memory (RAM), a magnetic or optical memory device (e.g., disk), a magnetic tape, a solid-state drive and the like. The memory medium may include flash memory cells, or other type memory, discrete EPROM or EEPROM, or the like. It is further noted that memory medium may be housed in a common controller housing with the one or more processors. In one embodiment, the memory medium may be located remotely with respect to the physical location of the one or more processors and controller. For instance, the one or more processors of controller may access a remote memory (e.g., server), accessible through a network (e.g., internet, intranet and the like).

[0103] As used throughout the present disclosure, the term “substrate” generally refers to a substrate formed of a semiconductor or non-semiconductor material (e.g., thin filmed glass, or the like). For example, a semiconductor or non-semiconductor material may include, but is not limited to, monocrystalline silicon, gallium arsenide, indium phosphide, or a glass material. A substrate may include one or more layers. For example, such layers may include, but are not limited to, a resist (including a photoresist), a dielectric material, a conductive material, and a semiconductive material. Many different types of such layers are known in the art, and the term sample as used herein is intended to encompass a substrate on which all types of such layers may be formed. One or more layers formed on a substrate may be patterned or un-patterned. For example, a substrate may include a plurality of dies, each having repeatable patterned features. Formation and processing of such layers of material may ultimately result in completed devices. Many different types of devices may be formed on a substrate, and the term substrate as used herein is intended to encompass a substrate on which any type of device known in the art is being fabricated. Further, for the purposes of the present disclosure, the term substrate and wafer should be interpreted as interchangeable. In addition, for the purposes of the present disclosure, the terms patterning device, mask, and reticle should be interpreted as interchangeable.

[0104] One skilled in the art will recognize that the herein described components operations, devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to

be representative of its class, and the non-inclusion of specific components, operations, devices, and objects should not be taken as limiting.

[0105] As used herein, directional terms such as “top,” “bottom,” “over,” “under,” “upper,” “upward,” “lower,” “down,” and “downward” are intended to provide relative positions for purposes of description, and are not intended to designate an absolute frame of reference. Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments.

[0106] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

[0107] The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected,” or “coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “couplable,” to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to physically mixable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

[0108] Furthermore, it is to be understood that the invention is defined by the appended claims. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” and the like). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). In those instances where a convention analogous to “at least one of A, B, or

C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

[0109] It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction, and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes. Furthermore, it is to be understood that the invention is defined by the appended claims.

Claims

1. A detector comprising: a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.
2. The detector of claim 1, wherein the dielectric layer is a silicon-based dielectric.
3. The detector of claim 2, wherein the dielectric layer is one of silicon oxide or silicon nitride.
4. The detector of claim 1, wherein the adhesion layer is a silicon-based Group IV material.
5. The detector of claim 4, wherein the adhesion layer is polycrystalline silicon.
6. The detector of claim 1, wherein a thickness of the adhesion layer is on an order of hundreds of nanometers or micrometers.
7. The detector of claim 6, wherein the thickness of the adhesion layer is on the order of hundreds of nanometers.
8. The detector of claim 1, wherein a width of the window is on an order of hundreds of micrometers, millimeters, tens of millimeters, or hundreds of millimeters.
9. The detector of claim 8, wherein the width of the window is on the order of hundreds of micrometers, millimeters, or tens of millimeters.
10. The detector of claim 9, wherein the width of the window is on the order of hundreds of micrometers or millimeters.
11. The detector of claim 1, wherein the boron concentration is at least 95 percent.
12. The detector of claim 11, wherein the boron concentration is at least 99 percent.
13. The detector of claim 1, wherein thicknesses of the upper boron layer and the lower boron layer are on an order of nanometers or tens of nanometers.
14. The detector of claim 13, wherein the thicknesses of the upper boron layer and the lower boron

layer are between 2 nm and 20 nm.

15. The detector of claim 14, wherein the thicknesses of the upper boron layer and the lower boron layer are between 2 nm and 10 nm.

16. The detector of claim 1, further comprising an anode and a cathode, wherein the anode is coupled to the lower boron layer, wherein the cathode is coupled to the silicon layer.

17. The detector of claim 1, further comprising a capping layer, wherein the capping layer is deposited on the lower boron layer.

18. The detector of claim 17, wherein the capping layer is an anti-reflective coating.

19. The detector of claim 1, wherein the detector is annular, wherein the detector defines an aperture through the silicon layer, the dielectric layer, the adhesion layer, the upper boron layer, and the epitaxial layer.

20. The detector of claim 19, wherein the detector is segmented into a plurality of channels configured to independently detect at least one of the electron or the photon.

21. The detector of claim 1, wherein the detector is one of a plurality of detectors of a detector array, wherein the silicon layer and the epitaxial layer are monolithic across the detector array.

22. The detector of claim 1, wherein the epitaxial layer is an n-type semiconductor which is lightly doped, wherein the silicon layer is heavily doped, wherein the p-n diode is a (p⁺)(n⁻)(n⁺) diode.

23. The detector of claim 1, wherein the adhesion layer and the upper boron layer are conductive and act as an electron shield for the dielectric layer.

24. A method comprising: depositing an epitaxial layer on a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; depositing a dielectric layer on the epitaxial layer; depositing an adhesion layer on the dielectric layer; etching a window through the adhesion layer and the dielectric layer to the epitaxial layer; and depositing boron layers, the boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode.

25. The method of claim 24, wherein the boron layers are deposited using a diborane species, wherein the diborane species does not laterally diffuse along the adhesion layer into the window.

26. The method of claim 24, further comprising removing the upper boron layer and the adhesion layer.

27. A scanning electron microscope comprising: a detector comprising: a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the epitaxial layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detector is configured to detect at least one of the electron or the photon.

28. The scanning electron microscope of claim 27, wherein the detector is annular, wherein the detector defines an aperture through the silicon layer, the dielectric layer, the adhesion layer, and the upper boron layer; and wherein the scanning electron microscope is configured to generate an

electron beam and pass the electron beam through the aperture.

29. A detector array comprising: an array of detectors, wherein the array of detectors comprise: a silicon layer, wherein the silicon layer is monocrystalline silicon and an n-type semiconductor; an epitaxial layer, wherein the epitaxial layer is deposited on the silicon layer; a dielectric layer, wherein the dielectric layer is deposited on the epitaxial layer; an adhesion layer, wherein the adhesion layer is deposited on the dielectric layer, wherein the dielectric layer and the adhesion layer define a window to the epitaxial layer; and boron layers including an upper boron layer and a lower boron layer, wherein the upper boron layer is deposited on the adhesion layer, wherein the lower boron layer is deposited on the silicon layer within the window, wherein the boron layers are p-type semiconductors, wherein the upper boron layer is separated from the lower boron layer by the dielectric layer, wherein the lower boron layer includes a boron concentration of at least 80 percent, wherein the lower boron layer is a pinhole-free film, wherein at least one of an electron or a photon is configured to penetrate the lower boron layer, wherein the lower boron layer, the epitaxial layer, and the silicon layer form a p-n diode by which the detectors are configured to detect at least one of the electron or the photon.
